



PAPER

Engineering anisotropic tissue analogues: harnessing synergistic potential of extrusion-based bioprinting and extracellular matrix-based bioink

To cite this article: Ashis Kumar Bera *et al* 2025 *Biofabrication* **17** 015003

View the [article online](#) for updates and enhancements.

You may also like

- [Chlorella-enriched hydrogels protect against myocardial damage and reactive oxygen species production in an *in vitro* ischemia/reperfusion model using cardiac spheroids](#)
Martine Tarsitano, Clara Liu Chung Ming, Lucia Bennar et al.
- [On-chip human lymph node stromal network for evaluating dendritic cell and T-cell trafficking](#)
Brian J Kwee, Mona Mansouri, Adovi Akue et al.
- [Standalone single- and bi-layered human skin 3D models supported by recombinant silk feature native spatial organization](#)
Savvini Gkouma, Nayanika Bhalla, Solène Frapard et al.



PAPER

Engineering anisotropic tissue analogues: harnessing synergistic potential of extrusion-based bioprinting and extracellular matrix-based bioink

RECEIVED
21 June 2024

REVISED
29 August 2024

ACCEPTED FOR PUBLICATION
15 October 2024

PUBLISHED
24 October 2024

Ashis Kumar Bera , Mohd Suhail Rizvi , Vijayasankar KN and Falguni Pati*

Department of Biomedical Engineering, Indian Institute of Technology Hyderabad, Kandi, Sangareddy, Telangana 502285, India

* Author to whom any correspondence should be addressed.

E-mail: falguni@bme.iith.ac.in

Keywords: bioprinting, alignment, decellularized extracellular matrix, shear stress, nozzle geometry, stretching

Supplementary material for this article is available [online](#)

Abstract

In the realm of tissue engineering, replicating the intricate alignment of cells and the extracellular matrix (ECM) found in native tissue has long been a challenge. Most recent studies have relied on complex multi-step processes to approximate native tissue alignment. To address this challenge, we introduce a novel, single-step method for constructing highly aligned fibrous structures within multi-modular three-dimensional conglomerates. Our approach harnesses the synergistic potential of extrusion-based bioprinting and the fibrillogenesis kinetics of collagen-rich decellularized ECM. We have identified three key parameters governing ECM microfiber alignment during extrusion-based bioprinting: applied shear stress, stretching or extensional force, and post-print deformation. By carefully manipulating these parameters, we have successfully created highly aligned fibrous structures within multi-modular three-dimensional conglomerates. Our technique offers an efficient solution and has been validated by computational modeling. Comprehensive analyses confirm the efficacy across various scenarios, including encapsulated, top-seeded, and migratory cells. Notably, we have demonstrated the versatility and effectiveness of our approach by bioprinting highly aligned cardiac tissue patches, which show further maturation evidenced by the expression of Troponin-T and Myo-D differentiation factor needed for contractility and myotube formation, respectively. In summary, our streamlined approach offers a robust solution for creating anisotropic tissue analogues with precise ECM organization.

1. Introduction

Tissues are not merely assemblies of cells and extracellular matrix (ECM); rather, they are intricately organized structures where cellular and matrix alignment play a fundamental role in defining their unique genetic to phenotypic expressions [1]. Alignment is crucial in many tissues to impart specific structural and mechanical properties necessary for their biological functions: think of aligned myoblasts and collagen in muscles facilitating uniaxial contraction [2], or bundles of aligned ECM in tendons transmitting force [3], and the precise spatial arrangement of collagen in the cornea ensuring transparency [4]. Aligned ECM fibers act as guides for embedded cells, promoting their alignment and contributing to dynamic

remodeling throughout the tissue's lifespan. To replicate the complex composition and alignment of ECM in tissue engineering, mimicking these features is paramount [5–7]. It is not just about assembling cells and matrix materials; it is about recreating the intricate architecture that underlies tissue functionality and behavior.

To mimic the complex architectural design, common methodologies such as surface patterning and topographical cues have been utilized to generate cellular alignment [8–14]. While effective in 2D cultures, these methods fall short of replicating realistic in vivo conditions. To address this limitation, researchers have strived to fabricate 3D scaffolds with controlled cell distribution and alignment for tissue engineering applications. Techniques such as laser

beam-patterned hydrogels [15, 16], micro-molding [17], and sacrificial template-based microchannel have been employed to align cells in the desired direction [18, 19]. Despite their success in fabricating miniature tissue constructs, challenges such as fabrication complexity, scaffold size, and low throughput hinder their clinical translation. In contrast, the advancement of additive manufacturing, particularly extrusion-based bioprinting, presents a promising approach to mimic tissue-specific architectural designs. Extrusion-based bioprinting allows for the precise deposition of cells and biomaterials at desired positions. Interestingly, the bioink extruded from the printing nozzle encounters significant shear stress from the nozzle wall, inducing alignment of microfibers present in the bioink in the desired direction. However, few studies have attempted to fabricate shear-induced alignment of synthetic and natural fiber-based biomaterials by extrusion-based bioprinting [20]. Alignment of biomaterials has primarily been achieved through the pre-fabrication of organ building blocks using gelatin and fibrinogen [21]. Another method involves compartmentalization of the nozzle to fabricate an aligned scaffold, where bioink is passed through several micro-compartments inside the extrusion nozzle, generating topological cues to facilitate scaffold alignment [22]. Mimicking tissue-specific architectural designs through extrusion-based bioprinting using fiber-based biomaterials offers a promising avenue to recreate the tissue-specific microenvironment. However, finding ideal biomaterials tailored to specific tissues remains a challenge due to our incomplete understanding of individual ECM proteins and their interactions within three-dimensional cellular environments [1, 23]. Collagen, the most abundant protein in native tissue, plays a pivotal role in influencing the properties of the entire tissue through its organization within the cellular microenvironment. Tissue-specific ECM possesses intrinsic characteristics that support seeded cells due to its specific composition and ultrastructure. Consequently, researchers have turned to tissue-specific ECMs as bioink for bioprinting [24]. Nonetheless, intense chemical treatment during the decellularization process compromises the ultrastructural integrity of tissues [25]. Thus, careful architectural orientation of ECM fibers is necessary to ensure optimal cellular activity when utilizing them as bioink for fabricating 3D structures. Achieving this balance between mimicking tissue-specific architecture and preserving ECM integrity is critical for advancing bioprinting techniques in tissue engineering.

In our study, we have tackled these challenges head-on by customizing nozzle geometry and optimizing various printing parameters to fabricate a microfiber-aligned scaffold, effectively guiding cellular growth and organization to support tissue

maturation. Through rigorous experimentation, we have identified three key parameters—shear stress, extensional force, and nozzle geometry—that significantly influence microfiber alignment in extrusion-based bioprinting. Our approach involved a thorough analysis of microfiber alignment in the biofabricated scaffold, utilizing both theoretical models and experimental techniques. We meticulously considered multiple printing parameters, including the rheological profile of the hydrogel, nozzle aspect ratio, and cell viability, to ensure precise control over microfiber alignment. Subsequently, we investigated cellular alignment within the microfiber-aligned scaffolds, employing various configurations such as encapsulated cells in a 3D environment, top-seeded cells, and cells allowed to migrate on the scaffold. This comprehensive characterization provided valuable insights into the efficacy of our approach in promoting cellular alignment and tissue maturation. As a proof of concept, we successfully demonstrated the utility of our technique by creating a tissue scaffold using aligned rat cardiomyocyte cells. These cells matured into myotubes, laying the foundation for the development of aligned cardiac tissue patches—an exciting prospect with significant implications for regenerative medicine and tissue engineering.

2. Materials and methods

2.1. Preparation and characterization of decellularized dermal matrix

Goat full-thickness skin from the dorsolateral flank was collected from nearby slaughterhouses and decellularized with our previously developed *NaCl*-based hypotonic and hypertonic solution method [26]. Briefly, the chopped dermis tissue was treated with 1 M *NaCl* hypertonic solution followed by 0.1 M *NaCl* hypotonic solution with continuous agitation at 320 RPM changing the solution every 4 h. The treatment with hypertonic and hypotonic was repeated for 2 d and maintained the tissue in a hypertonic solution during night. Then, the hypertonic and hypotonic solution treatment was repeated for another 2 d changing the solutions every 6 h. After 4 d of *NaCl* treatment, the tissue was washed with 3% H_2O_2 for 15 min. Then the tissues were washed with deionized water for 2 d by changing the water every 6 h. After completing deionizing water wash the tissues are kept at $-80\text{ }^{\circ}\text{C}$ prior to lyophilization. Then the lyophilized tissue was digested by papain to characterize remaining structural protein collagen by hydroxyproline method and functional protein as glycosaminoglycan by DMMB based sGAG quantifying method. Also evaluate the degree of decellularization by quantifying remaining DNA in the tissue matrix by picogreen assay. Then we have prepared decellularized dermal extracellular matrix (dECM) hydrogel by digesting the lyophilized powder in 0.5 M acetic

acid solution containing 10% pepsin. All the characterization, bioprinting and cellular study were done by 2% hydrogel only.

2.2. Rheological analysis of dECM hydrogel

The rheological profile of dECM hydrogel has been performed by Malvern Kinexus+ rheometer. Hydrogel used for rheological analysis was 2% dECM solution. The rheological sequence has been chosen for the stability of hydrogel as well as requirements of specific characteristics for extrusion based bioprinting. We have performed amplitude sweep from 0.1% to 100% to evaluate viscoelastic behavior of hydrogel in different shear strain. The shear thinning property performed by exposing low to very high strain rate and shear recovery properties of hydrogel has been measured by exposure to high shear rate followed by low shear rate. The applied shear rate and stress at the tip of the nozzle has been calculated by thixotropic rebuilt analysis in respect to nozzle diameter, volumetric flow rate in different set of print speed.

2.3. Customization of stainless-steel nozzle

We have customized the widely used stainless-steel dispensing nozzle used for extrusion based bioprinting. We have chosen a 23-gauge needle for customization to desired shape and used the same nozzle without modifying as a control for 3D bioprinting. We captured images of customized nozzles by stereo microscope and evaluated the shape & geometry and designed by CAD–CAM software to perform flow simulation.

2.4. COMSOL fluid flow simulation of extrusion-based bioprinting

In the customized nozzle the flow behavior of bioink with applied pneumatic pressure was simulated by COMSOL multiphysics 6.2 software. Where the flow behavior as well as pressure distribution was analyzed in different position of our customized nozzle. The nozzle boundary was considered as a no-slip boundary and simulate with the viscosity and density of our hydrogel's property.

2.5. Cells and bioink

To analyze cellular alignment and fabricate a scaffold, we procured NIH mouse embryonic fibroblast cell line and H9C2 cardiomyocyte from NCCS (Pune, India) for this study. We maintained this cell line in DMEM high glucose with L-Glutamine, and Pyruvate for continuous culture and used them for experiment with 70% of confluency. We have prepared 2% of dECM hydrogel from lyophilized tissue and use the same for all the experiments. For bioink preparation we added 10% of 10X media and FBS to pH adjusted dECM hydrogel to maintain the osmolality as well as to support cells during printing and subsequent

gelation. We used 6×10^6 cells ml^{-1} of hydrogel for bioprinting.

2.6. 3D bioprinting with dECM hydrogel by different geometry nozzle

We have chosen four different sizes and shapes of nozzles to extrude 2% dECM hydrogel by extrusion based bioprinting. Here, we have selected 23 Gauge stainless steel nozzle having inner diameter of $330 \mu\text{m}$. All the structures were printed at 15°C with a 3 cc cartridge by NBIL (Bengaluru, India) bioprinter. To generate a tissue construct with aligned microfiber, we have optimized the printing parameters to apply different shear stress on dECM hydrogel. First, we have designed a rectangle sheet ($15 \times 10 \text{ mm}$) by computer-aided design (CAD) software and printed on 35 mm petri dish. We have chosen the printing speed from 100 mm min^{-1} to 5000 mm min^{-1} with respective pneumatic pressure. For every point of print speed, we printed the same lattice structure with 30% of infill percent grid structure and analyzed print accuracy by bright field microscope imaging. The applied pneumatic pressure was found out by optimizing the print accuracy by considering a $330 \pm 10\%$ -line width ($297\text{--}363 \mu\text{m}$) with respect to $330 \mu\text{m}$ nozzle diameter. With optimized pneumatic pressure we have printed the final lattice structure with 100% infill while considering 15%-line overlapping from each side. In all the printed constructs, the volumetric flow rate has been calculated by weighing the petri dish before and after printing and time taken for printing such volume. After printing, the construct has been kept at 37°C incubator with 5% CO_2 for 40 min.

2.7. Characterization of rheological profile with printing parameters

The second step of optimization of printing parameters is the characterization of hydrogel properties when it passes through the printing nozzles as well as after deposition to the bed. The rheological profile of hydrogel corresponding to printing parameters was analyzed by rheometer (Kinexus+, Malvern Instruments, UK). We have calculated volumetric flow rate at different print speeds and calculated shear stress and viscosity from equilibrium flow curve of hydrogel. First, we calculated the power law index from shear rate v/s viscosity plot and calculated the exerted shear strain in specific volumetric flow rate at specific print speed. We simulated the extrusion of hydrogel at constant diameter of nozzles, volumetric flow rate and power law index to get the applied shear strain in that condition from the thixotropy analysis. Then we have calculated the exerted shear stress and viscosity at the tip of the nozzle for each number of experiments from the strain v/s viscosity plot. As we have tried printing constructs at different

flow rates, effectively at different shear rate, the hydrogel's rebuild property has been measured, which is the time taken to regain its elastic modulus after extruding from a nozzle as it directly influences the materials' printability.

2.8. Characterization of spreading behavior of post printing hydrogel by Hi-Speed camera

To examine the spreading behavior of our bioink after extruding from the nozzle, we captured videos by Hi-Speed camera (Phantom VEO 710L, Phantom, USA) and analyzed it over time. We first printed a single line with both the nozzles and captured videos with 1000 FPS (frames per second) from both front view and side view. The images are extracted by Phantom cine view software and analyze its spreading behavior and calculate the line overlap required to get a sheet like construct with 100% infill.

2.9. Computational model of collagen fiber alignment during extrusion based bioprinting

For computational model of the bioprinting process with our hydrogel, we model the collagen fibers as rigid rods of mean length l , and diameter a , and consider them to be arranged in an orientationally isotropic fashion in the cylindrical nozzle of length L . The individual collagen fibers are assumed to follow the flow velocity profile with minimal hydrodynamic interactions among them, thanks to their high density resulting in screening effects [27]. We consider nozzle geometry to be cylindrical with an elliptical cross-section with major and minor axes l_x and l_y , respectively. We characterize the shape of the nozzle by its aspect ratio which is defined as $\alpha = l_x/l_y$. The lower end of the nozzle is kept at a height H from the substrate on which the printed structure is collected. The nozzle is moved during the printing process in a manner that the major axis of the elliptical cross-section remains normal to the nozzle axis and the printing direction (see figure below). The flow velocity of the bioink in the nozzle (relative to the nozzle velocity) is assumed to be of parabolic profile and is specified by.

$$v_n(x, y) = v_0 (1 - r^2(x, y))$$

where,

$$r(x, y) = \sqrt{\frac{x^2}{l_x^2} + \frac{y^2}{l_y^2}}$$

is the normalized distance of a point in the cross-section of the nozzle from its center. It must be noted that this description of the flow velocity ensures no-slip condition at the nozzle wall. Further, the flow of the bioink inside the nozzle is taken to be fully developed and, therefore, flow velocity does not depend on the z coordinate. As mentioned earlier the

collagen fibers are assumed to follow this flow velocity inside the nozzle. The interaction between individual collagen fibers and the nozzle wall is taken to be of hard repulsion to make sure fibers do not cross the nozzle wall. The interaction potential between a fiber end and nozzle wall is, therefore, given by

$$U(d) = \begin{cases} \infty, & \text{if } d < 0 \\ 0, & \text{otherwise} \end{cases}$$

where d is the radial distance of the fiber end from the nozzle wall.

At the end of each simulation, the orientation of each collagen fiber is calculated in the $x-y$ plane (similar to the image analysis in experiments) with the help of the coordinates of its two ends. This calculation gives the orientation of each fiber $\theta \in (-\pi, \pi)$ which are plotted as histogram plots in the polar plane. Further, the average orientation order parameter is also calculated as

$$P = \langle \cos(\theta_i) \rangle$$

where θ_i is the angle of the fiber relative to the printing direction, and $\langle \cdot \rangle$ represents the ensemble average. Details has been described in supplementary document 2.

2.10. Cells viability assay

The cell encapsulated bioprinted construct was washed twice in PBS and then incubated with Live/DEAD (Invitrogen) solution containing Calcein-AM (2×10^{-6} M) and ethidium-homodimer-1 (4×10^{-6} M) for 30 min at 37 °C and 5% CO₂. Next, we have checked the cell viability at different shear stress after 1 d of printing. After incubation, the images were captured by a confocal microscope and quantified the cell viability. Calcein AM was taken up by live cells and converted to green-fluorescent metabolite fluorescein by esterase enzyme and ethidium-homodimer-1 was taken up by cells and fluoresce red when the cell membranes are fragile, which indicates dead cells.

2.11. Quantify alignment of sheet microfiber by Scanning Electron Microscopy (SEM) analysis

To check the surface structure and quantify the fiber alignment on the bioprinted constructs, samples have been fixed and analyzed by SEM. The bioprinted constructs were fixed with 4% paraformaldehyde (PFA) for 45 min at room temperature and then washed with PBS three times. Then the constructs were frozen gradually from 4 °C to -20 °C and finally at -80 °C for another 2 d prior to freeze drying. The lyophilized samples were coated with gold sputtering for 45 s and captured images with SEM. The images were captured by minimum magnification (500x) to very high magnification (50 000x) to analyze the gross alignment as well as alignment of collagen and other microfibers.

Table 1. Microfiber alignment with different printing parameters with respect to hydrogel's rheological profile.

Print Speed (mm min ⁻¹)	Resultant Vector (a.u.)				Shear Stress (Pa)	Shear Rate (s ⁻¹)	Shear Viscosity (mPa s)
	NAR* 1	NAR* 1.4	NAR* 2.8	NAR* 4.2			
100	0.68	0.76	0.86	0.89	36.96	17.4	2130
500	0.77	0.82	0.88	0.90	57.11	87.2	657
1000	0.83	0.85	0.89	0.91	68.86	174.4	396.96
2000	0.88	0.89	0.91	0.92	82.74	344.21	241.82
3000	0.90	0.91	0.92	0.92	92.79	523.21	178.2

NAR*: Nozzle Aspect Ratio.

2.12. Cells migration assay on top of aligned ddECM sheet

Cells migration was estimated on the bioprinted construct, where the effect of microfiber alignment on cellular migration was evaluated. We have printed all the constructs at different print speeds with dimensions of 15 × 10 mm having 100% infill. Along with the constructs a single line was also printed having 200 μm of width and 40 mm of length. The printed constructs were kept for gelation at 37 °C with 5% CO₂ for 40 min. After gelation the constructs were submerged in PBS overnight for swelling and stability studies. On the next day, PBS was discarded, and the single printed line was placed on top of the printed sheets for all the constructs. Then, NIH 3T3 mouse fibroblast cells were top seeded on the constructs and cultured at 37 °C with 5% CO₂ to reach confluence. After the cells became confluent (more than 95%) the single printed line was taken off from the construct with a micropipette without disturbing the construct. The place where the single printed line was there becoming a cell exclusion zone and the cell migration to that zone was evaluated by bright field microscopy.

2.13. Immunostaining and quantifying cells alignment

To prepare samples for immunostaining, the bioprinted constructs were rinsed with PBS and fixed in 4% PFA solution for 45 min at room temperature. The fixed samples were washed with PBS three times and treated with 1% BSA and 0.1% Triton-X solution for 30 mins. Then the constructs were washed twice by PBS and incubated with Rhodamine Phalloidin (R415, Invitrogen) at 1:200 dilutions for 30 min at 37 °C. Following this incubation, the constructs were washed three times by PBS and treated with Hoechst (H1398, Invitrogen) at 1:3000 dilutions for 10 min at room temperature. After that the constructs were washed with PBS and images were captured by Confocal Microscope (SP8 Lightning Confocal Microscope, Leica).

2.14. Quantifying ECM microfiber and cellular alignment

To quantify the alignment of ddECM fiber, encapsulated cells, and migrated cells on top of the printed constructs, we captured images by SEM and confocal microscope and analyzed them by Image J and Cir r plugin in MATLAB. To identify the ddECM fiber, the SEM images of different magnifications have been chosen and analyzed using ImageJ plugin directionality. The exported data is further analyzed by MATLAB to extract the resultant vector; that represents 0 and 1 matrix alignment ranges. In the same way, Phalloidin staining images for encapsulated cells and top-seeded migrated cells were analyzed to quantify the alignment.

2.15. Gene expression analysis

Total RNA from the bioprinted scaffold was isolated using the standard Trizol method at different time points according to experiments using Ambion TRIzol™ Reagent from Thermo Fisher Scientific. The obtained total RNA was dissolved in DEPC (Diethylpyrocarbonate) (HiMedia) treated water, and the A260/A280 ratio was calculated using the NanoDrop™ 2000 Spectrophotometer. Only samples with an A260/A280 ratio of 1.8–2.0 were included for cDNA preparation. Samples with 25 ng μl⁻¹ of RNA concentration were reverse transcribed to cDNA (Takara Bio's PrimeScript 1st strand cDNA Synthesis Kit) according to manufacturer protocol. For the final reaction mixture, 20 μl of reaction mix was made using Takara Bio's TB Green Premix Ex Taq II (Tli RNase H Plus), wherein both forward and reverse primer had a final concentration of 1 pmol μl⁻¹ in the total reaction mix. Forward and reverse primer sequences have been included in supplementary information (SD1 table 1). We used Qiagen Rotor Gen Q PCR Machine, and the PCR cycles had a denaturation phase set at 95 °C for 20 s, followed by an annealing phase at 60 °C for 20 s and an extension phase at 72 °C for 30 s with a total of 40 reaction cycles per run. At the end of each reaction, Cq values for each gene of interest were calculated

in triplicates and normalized to Cq values for the housekeeping gene β -actin. We have represented the relative gene expression on day 1 and day 7 in compare to housekeeping gene β -actin.

2.16. Statistical analysis

All the quantified data was presented as Mean $\pm$ SEM unless stated otherwise. To evaluate significant differences, we performed a student t-test and defined significance as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

3. Results and discussion

In this study, we have demonstrated the fabrication of a microfiber-aligned tissue structure using extrusion-based 3D bioprinting. Our methodology revolves around applying shear stress via the nozzle wall, exerting extensional force through nozzle movement, and controlling the spreading of a fiber-rich biomaterial formulation by customizing nozzle geometry. This enables us to align the fibers along the printing direction, effectively harnessing the capabilities of extrusion-based bioprinting for precise tissue engineering. One of the primary challenges in bioprinting is selecting a hydrogel that not only supports cellular growth but also mimics physiological ultrastructure. dECM possesses tissue-specific extracellular matrices that provide excellent support for seeded cells. Consequently, dECM-based bioinks have been widely utilized for fabricating tissue analogs using extrusion-based bioprinting techniques [24]. However, due to their low viscosity and irregular organization of ECM fibers, achieving aligned 3D structures with dECM bioinks has proven to be challenging.

In our study, we tackled the challenges of architectural design at the cellular level using an extrusion-based bioprinting technique. Specifically, we focused on optimizing various printing parameters to ensure the alignment of ECM microfibers along the printing path. In conventional bioprinting, the stainless-steel round-shaped nozzle is commonly employed for extruding low-viscosity materials to achieve high-resolution printed constructs. However, with the round-shaped nozzle, the bioink experiences non-uniform shear stress, gradually decreasing from the periphery to the center (figure S1(a)). Furthermore, the cylindrical extruded filament makes minimal contact with the bed, leading to spreading due to gravitational forces, resulting in transversely distorted filaments. This spreading of the hydrogel minimizes microfiber alignment in conventional printing with a round-shaped nozzle.

To address these challenges, we innovatively customized the stainless-steel round-shaped nozzle to a semi-elliptical shape. This modification ensures the application of uniform shear stress to the hydrogel and facilitates a broader area of contact with

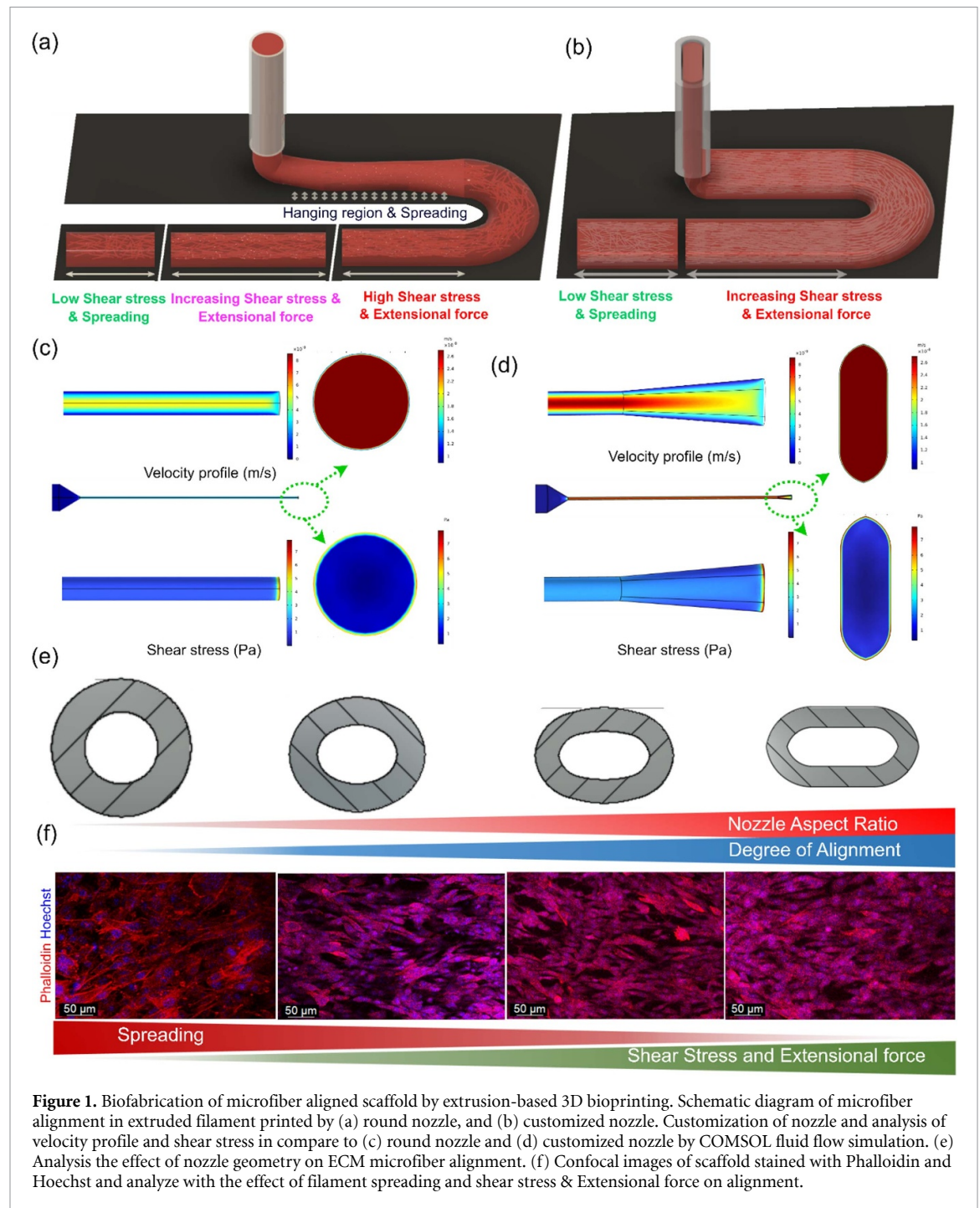
the bed after extrusion from the nozzle, minimizing hanging regions and preventing further distortion (figures 1(a), (b) and S1(b)). This customization significantly enhances the precision and alignment of microfibers during the bioprinting process, thereby improving the overall quality and fidelity of the printed constructs.

Furthermore, we have demonstrated that increasing print speed results in better alignment of ECM fibers without compromising cell viability. Numerical simulation results have corroborated these findings, revealing differences in streamline flow, shear stress, and extensional force applied on the hydrogel by nozzle movement for different shapes of nozzles with the same length and diameter (figures 1(c), (d) and S(d2)). Through the utilization of our customized nozzles, we have successfully fabricated scaffolds with aligned ECM fibers and cells. Importantly, we have shown that the alignment can be modulated according to design specifications and to varying degrees without detrimental effects on cell viability. Nozzle aspect ratio, filament spreading, shear stress, and extensional force—all of which were meticulously examined using high-speed camera videography and numerical simulation—play crucial roles in contributing to the alignment of ECM microfibers and cells (figures 1(e) and (f)).

The ultimate objective of biofabrication is to achieve alignment of cells within internal microfibers, mimicking a hierarchical ultrastructure akin to native fibrillar tissue. The biofabrication method developed in this study facilitates the fabrication of constructs with superior ultrastructural organization in a simple and cost-effective manner, without the need for special tools. This represents a significant advancement in tissue engineering, offering the potential to create functional tissue constructs with enhanced structural integrity and functionality.

3.1. Design of printing parameters through hydrogel's rheological profile

Our strategy for fabricating a scaffold with aligned microfibers revolves around two key elements: the design of a collagen microfiber-rich hydrogel and the extrusion-based biofabrication process. To achieve this, we selected decellularized dermal tissue from caprine as the matrix for tissue construct fabrication. We utilized our previously developed NaCl-based protocol for decellularizing goat dermal tissue, followed by an evaluation of the resultant matrix's suitability for tissue engineering applications [26]. Biochemical analysis revealed that less than 3% of DNA was retained in the decellularized matrix, while a significant amount of collagen ($\sim 60\%$) and sulfated glycosaminoglycan ($\sim 80\%$) remained (figure S2(a)). This retention of collagen and sulfated glycosaminoglycan is crucial for supporting cellular functionality [25]. In dermal tissue, Type-I collagen



predominates, imparting a fibrillar nature to the matrix. Leveraging this, we utilized Type-I collagen as the primary matrix fiber to align by modulating process parameters via extrusion-based 3D bioprinting techniques.

To fabricate a scaffold with aligned microfibers, we meticulously designed a set of printing parameters tailored to the rheological properties of the hydrogel. Previous studies have highlighted that applying shear stress to fiber-based hydrogels can align the fibers along the printing direction [21, 22, 28]. However, it is crucial to avoid excessively high shear stress

levels that could harm encapsulated cells. Therefore, selecting an extrusion rate that generates moderate shear stress, sufficient for aligning ECM fibers and seeded cells without compromising cellular viability, is paramount.

The rheological profile, comprising a range of physicochemical properties characterizing the hydrogel's deformation and flow behavior, has been instrumental in controlling microfiber alignment via extrusion-based bioprinting (figures S2(b)–(d) [29]. To regulate the flow rate and thus the shear stress, we directly adjusted the applied pneumatic pressure and

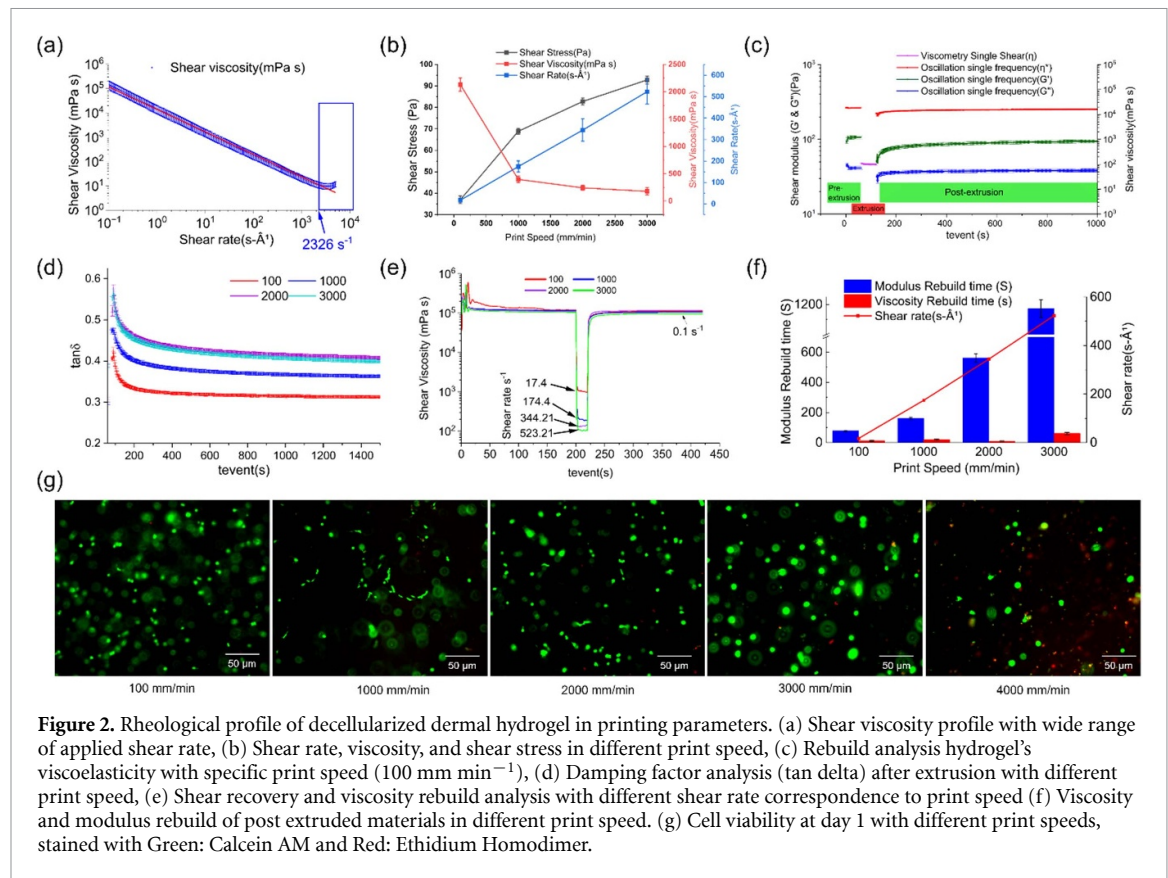


Figure 2. Rheological profile of decellularized dermal hydrogel in printing parameters. (a) Shear viscosity profile with wide range of applied shear rate, (b) Shear rate, viscosity, and shear stress in different print speed, (c) Rebuild analysis hydrogel's viscoelasticity with specific print speed ($100 mm min^{-1}$), (d) Damping factor analysis (tan delta) after extrusion with different print speed, (e) Shear recovery and viscosity rebuild analysis with different shear rate correspondence to print speed (f) Viscosity and modulus rebuild of post extruded materials in different print speed. (g) Cell viability at day 1 with different print speeds, stained with Green: Calcein AM and Red: Ethidium Homodimer.

print speed to evaluate their effects on alignment. We conducted a comprehensive analysis to determine the acceptable range of shear rate for our bioink, utilizing a complete range of oscillation shear rate sweeps. Our hydrogel exhibited a loss of shear thinning ability at applied shear rates greater than $2326 s^{-1}$, behaving like a Newtonian fluid with no viscosity variation with increasing shear rate (figure 2(a)).

Next, we printed scaffolds with 100% infill at different print speeds, along with corresponding pneumatic pressures, and calculated the applied shear rate and shear stress. To ensure optimal line width ($330 \pm 33 \mu m$, considering 10% spreading) consistent with the nozzle diameter ($330 \mu m$), we optimized the required pressure for each print speed. To ascertain the range of stress levels for scaffold fabrication, we varied the print speed and calculated the applied shear stress on the nozzle tip for each speed. Initially, we determined the volumetric flow rate experimentally and assessed the shear rate using the Malvern Kinexus+ rheometer. Subsequently, we calculated viscosity using shear rate versus viscosity plots (figure 2(b)). This enabled us to pinpoint the upper limit of print speed ($3000 mm min^{-1}$), beyond which cellular viability significantly decreases (figures 2(g) and S3). Thus, we selected a maximum print speed of $3000 mm min^{-1}$ for further analysis of printing accuracy and microfiber alignment.

While the rheological profile from strain sweeps defined the acceptable range of shear rate for the maximum printing speed, shape fidelity emerged as a critical parameter for printability, directly correlating with the applied shear rate [29]. Shape fidelity, referring to the ability of the extruded filament or whole construct to retain its shape according to a CAD model, was another crucial consideration. To evaluate the shape fidelity of printed constructs, we analyzed the time required for the hydrogel to regain 90% of its shear modulus after extrusion at each printing speed using oscillation rheology with the Malvern Kinexus+ Rheometer. We measured the complex shear viscosity in oscillatory single frequency (pre-extrusion), viscometry single shear (extrusion time), and complex shear viscosity in oscillation single frequency (post-extrusion) (figure 2(c)) for $100 mm min^{-1}$.

Our analysis revealed that the rebuild time increased exponentially with increasing printing speed and shear rate, resulting in decreased shape fidelity and compromised printing accuracy. At different single shear rate exposures, we observed that our hydrogel's shear modulus and shear viscosity exhibited different regaining properties. Therefore, the ratio of G''/G' , or damping factor (tan delta), emerged as a crucial factor influencing shape retention properties. We evaluated tan delta for each printing parameter and found that it increased with

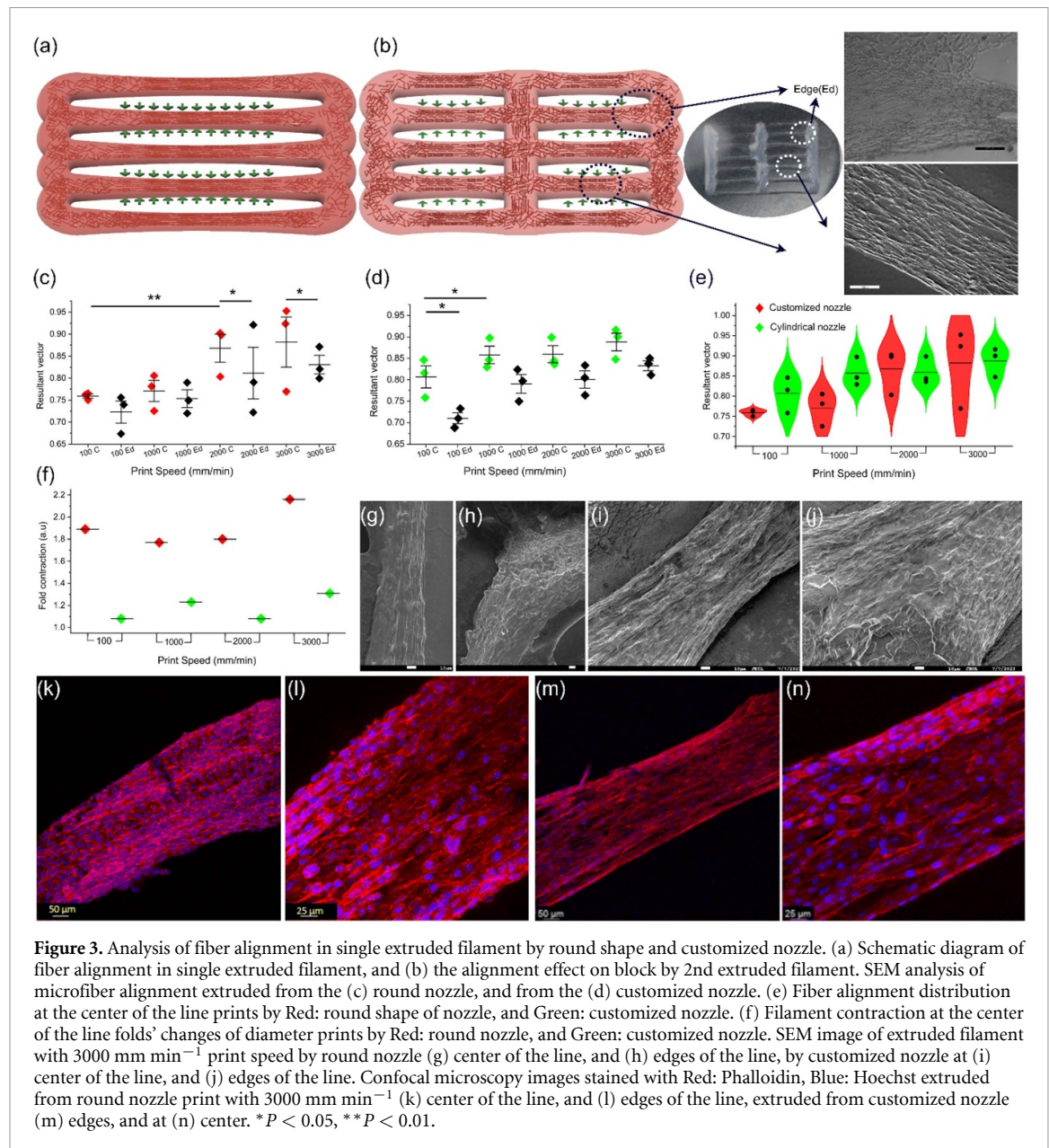


Figure 3. Analysis of fiber alignment in single extruded filament by round shape and customized nozzle. (a) Schematic diagram of fiber alignment in single extruded filament, and (b) the alignment effect on block by 2nd extruded filament. SEM analysis of microfiber alignment extruded from the (c) round nozzle, and from the (d) customized nozzle. (e) Fiber alignment distribution at the center of the line prints by Red: round shape of nozzle, and Green: customized nozzle. (f) Filament contraction at the center of the line folds' changes of diameter prints by Red: round nozzle, and Green: customized nozzle. SEM image of extruded filament with 3000 mm min⁻¹ print speed by round nozzle (g) center of the line, and (h) edges of the line, by customized nozzle at (i) center of the line, and (j) edges of the line. Confocal microscopy images stained with Red: Phalloidin, Blue: Hoechst extruded from round nozzle print with 3000 mm min⁻¹ (k) center of the line, and (l) edges of the line, extruded from customized nozzle (m) edges, and at (n) center. * $P < 0.05$, ** $P < 0.01$.

increasing print speed, indicating a decrease in shape retention with rising shear rate (figure 2(d)).

To further confirm the shape retention property, we performed oscillation shear recovery at different shear rates and compared the time taken for modulus rebuild. Our findings demonstrated that shear recovery to modulus rebuild was much faster at lower shear rates. Specifically, shear viscosity regained 90% within 29 s at shear rates of 344.21 S⁻¹ or less, compared to 60 s at 523.21 S⁻¹ (3000 mm min⁻¹ print speed) (figures 2(e) and (f)).

3.2. Directing cellular organization in ddECM extruded filament

To initiate the fabrication of a scaffold with aligned microfibers, we began by printing a rectilinear lattice structure and scrutinizing its alignment using the applied printing parameters. Previous studies have

indicated that microfibers in the bioink can align with the printing direction due to the applied shear stress from the nozzle wall [21, 30]. Utilizing a commonly used round-shaped opening 23-gauge needle, we printed a line at a print speed of 2000 mm min⁻¹ with corresponding pneumatic pressure (figure S4(a)).

After a 7 day culture period, we observed a significant reduction in line width towards the center compared to its edges, resulting in better cellular alignment at the center than at the edges (figures 3(a) and S4(b)–(g)). The resultant vector of cellular alignment at the center and edges was calculated to be 0.7259 ± 0.02 and 0.5840 ± 0.015 , respectively (figure S4(h)). This asymmetric alignment phenomenon can be attributed to the contraction of the line at the center during culture. Since the line was freely floating without attachment to the tissue culture plate, compression in the cross-sectional area likely occurred

(figures S4(b)–(g)). When microfibers were aligned towards the end of the line, they were confined by the perimeter backbone, preventing contraction, and leading to reduced alignment.

It is evident that the nozzle wall and nozzle movement can create shear stress and extensional force, resulting in a minimal degree of alignment. This preliminary observation underscores the importance of optimizing printing parameters and scaffold design to achieve consistent and robust microfiber alignment for advanced tissue engineering applications.

For further evaluation, we undertook a comparative analysis between the customized extrusion nozzle and the standard round-shaped nozzle to elucidate the pattern of applied shear stress, the hydrogel spreading during deposition, and subsequent contractions of the printed line in cross-section (figure S5). Additionally, we delved into the hydrogel flow behavior inside the nozzle at various locations using COMSOL Multiphysics fluid flow simulation (figure S1). Our analysis revealed significant differences in the velocity gradient and wall shear stress, particularly at the compressed zone (semielliptical) compared to the round part of the nozzle. Notably, the maximum volume of hydrogel experienced uniform shear stress within the compressed zone. Moreover, within the nozzle, a velocity gradient and corresponding shear stress at the wall were observed, potentially modulating the hydrogel microfiber orientation along the nozzle's path.

This comprehensive analysis sheds light on how nozzle geometry can influence the distribution of shear stress and hydrogel flow behavior during the extrusion process. Such insights are invaluable for optimizing printing parameters and nozzle design to achieve precise microfiber alignment, advancing the capabilities of extrusion-based bioprinting for tissue engineering applications.

To evaluate ECM microfiber alignment under different printing parameters, we printed a lattice structure encapsulating NIH 3T3 mouse fibroblast cells using both the standard round-shaped nozzle and the customized nozzle, with varied print speeds ranging from 100–3000 mm min^{−1} and corresponding pneumatic pressures (figure S6). After culturing the scaffold for 7 d, fixation with formalin, and processing through serial dehydration and freezing, we examined the alignment of ECM microfibers under a SEM.

Our observations indicate a significant influence of print speed on ECM microfiber alignment, with alignment degree increasing as print speed rises, as evidenced by SEM images. However, we noted that microfibers are not uniformly aligned across the scaffold, with greater alignment observed at the center compared to the edges (figures 3(b) and (c)). Interestingly, with our customized nozzle, microfibers exhibited better alignment at the center compared to the edges, particularly at low print

speeds (100 mm min^{−1}), resembling the alignment pattern seen with the round-shaped nozzle. However, at higher print speeds (1000 mm min^{−1} or more), the difference in alignment between the center and edges diminished with the customized nozzle (figure 3(d)).

Our findings indicate that increasing microfiber alignment with print speed is attributed to the applied extensional force at higher speeds, leading to stretching of ECM microfibers along the printing direction and promoting alignment. Notably, at a print speed of 2000 mm min^{−1}, microfibers reached their maximum extension potential, with no further extension observed beyond this speed. The maximum alignment of embedded microfibers was achieved at 2000 mm min^{−1} print speed for the round-shaped nozzle. However, with the customized nozzle, a print speed of only 1000 mm min^{−1} was sufficient to achieve a similar degree of microfiber alignment, likely due to comparable extensional force.

Despite both nozzles exerting similar shear stress and extensional force at a given print speed, the alignment was notably lower with the round-shaped nozzle. This discrepancy may be attributed to filament spreading and distortion post-extrusion, affecting alignment outcomes. These insights underscore the importance of nozzle design in influencing microfiber alignment during the bioprinting process.

To delve deeper into the effect of filament spreading on microfiber alignment, we conducted an analysis of microfiber alignment at different positions within the center of the line (figures S7 and S8). Our findings revealed non-uniform alignment of fibers within the line, with better alignment observed at the edges compared to the mid area, and a wide range of alignment distribution with the round-shaped nozzle (figure 3(e), red). In contrast, alignment was nearly uniform for all filaments printed with our customized nozzle, with minimal or insignificant filament spreading post-deposition (figure 3(e), green and figures S7, S8).

Further analysis involved measuring filament diameter at different positions along the line, revealing a higher degree of alignment at the center (defined as C) compared to the edges (defined as R and L), primarily due to filament contraction (figure S8). The average contraction of filaments at the center was more than 1.8-fold greater than at the edges for the round-shaped nozzle, whereas it was only 1.2-fold for our customized nozzle (figure 3(f)).

Visual examination of filaments extruded from the round-shaped nozzle at the center and edges of the line at a print speed of 3000 mm min^{−1} (figures 3(g) and (h)), and similarly for the customized nozzle (figures 3(i) and (j)), further underscored the differences in alignment behavior. Additionally, the alignment of encapsulated cells within extruded filaments from both nozzles was cross-checked using immunofluorescence microscopic imaging with Phalloidin

and Hoechst staining (figures 3(k)–(n)). The results revealed significantly higher cellular alignment with the customized nozzle compared to the round-shaped nozzle (figure S9).

These comprehensive analyses provide further insights into the impact of filament spreading on microfiber alignment and cellular organization, highlighting the superior performance of the customized nozzle in achieving uniform alignment and enhancing cellular alignment within the printed scaffolds.

3.3. Evaluation the effect of nozzle geometry on microfiber alignment by videography and mathematical modeling

To assess the impact of nozzle geometry on microfiber and cellular alignment, we conducted videography using a Hi-Speed Camera (Phantom VEO 710L) capturing footage at 1000 FPS. This allowed us to closely examine the deposition and spreading behavior of the extruded filament. Our analysis revealed distinct differences between the round-shaped and customized nozzles. For the round-shaped nozzle, the bioink made contact with the bed at an angle of $54.26 \pm 7^\circ$, whereas with our customized nozzle, this angle increased to $86.3 \pm 8.3^\circ$ (figures 4(a), (b) and S10). This difference in angle suggests variations in the initial positioning of the deposited filament, which can affect subsequent alignment and organization within the scaffold.

Another set of videography enabled us to analyze the spreading behavior of the hydrogel after extrusion from both nozzles. We observed that the extruded filaments spread over time, leading to an increase in width. Specifically, for the round-shaped nozzle, the width increased from $157 \mu\text{m}$ at '0' seconds to $373 \mu\text{m}$ at 4 s, indicating significant spreading. In contrast, with our customized nozzle, the width increased from $400 \mu\text{m}$ at '0' seconds to $626 \mu\text{m}$ at 0.1 s before stabilizing for the next 4 s (figures 4(c)–(e) and S11). This difference in spreading behavior highlights the influence of nozzle geometry on filament deposition and subsequent scaffold formation. These findings underscore the importance of nozzle design in controlling the deposition and spreading of bioink during the bioprinting process, ultimately impacting microfiber alignment and cellular organization within the fabricated scaffold.

To gain deeper insights into the influence of nozzle geometries and printing speed on microfiber alignment, we conducted fluid flow modeling using various types of nozzle geometries. In our computational model, we assumed collagen fibers to be rigid rods with a mean length ' l ' and diameter ' a ', arranged in an orientationally isotropic fashion within the cylindrical nozzle of length ' L '. These fibers were assumed to follow the flow velocity profile within the nozzle, with minimal hydrodynamic interactions among them due to their high density resulting

in screening effects [27]. For the simulation results presented, we set $l/a = 50$, with nozzle geometries characterized by parameters $\alpha = 1, 1.4, 2.8$, and 4.2 . Each simulation was initialized with 10^6 collagen fibers inside the nozzle. The orientations of the fibers within the nozzle were taken from a uniform distribution in three dimensions. We started the simulation by moving the nozzle with a constant velocity v_p , while the fibers followed the flow within the nozzle and were subsequently extruded and settled. The values of v_p used were 100, 500, 1000, 2000, and 3000 mm min^{-1} . Each simulation was terminated once 95% of the fibers had been extruded from the nozzle (figure 4(f)). At the end of each simulation, the orientation of each collagen fiber was calculated in the xy plane using the coordinates of its two ends. This calculation provided the orientation of each fiber, which was then plotted as histogram plots in the polar plane (figure 4(g)). Additionally, the average orientation order parameter relative to the printing direction was calculated and represents the ensemble average (figure 4(h)).

By analyzing the results of these simulations, we aimed to better understand how nozzle geometries affect microfiber alignment and validate our experimental findings. This computational approach provides valuable insights into the underlying mechanisms governing microfiber behavior during the bioprinting process.

3.4. Characterization of microfiber and cellular alignment in gradient infill scaffold

Despite successful fabrication of single filaments with aligned ECM microfibers and seeded cells using extrusion-based 3D bioprinting, we encountered significant challenges when attempting to print sheet-like scaffolds with the same parameters [16, 18, 22, 27–34]. To investigate this discrepancy further, we designed a scaffold measuring $15 \times 10 \text{ mm}^2$ with gradient infill and printed it using our customized nozzle. After culturing the scaffold for 7 d, we stained it with Phalloidin and Hoechst and captured images at different locations using a confocal microscope to assess alignment. Surprisingly, while the cells remained aligned throughout the scaffold, their alignment differed markedly between the single filament and infill regions. Specifically, we observed uniaxial alignment of cells in a single line, consistent with the direction of nozzle movement. However, in the infill region, cell alignment appeared random, indicating a loss of alignment compared to the single filament structure. This discrepancy highlights the complexity of achieving consistent microfiber alignment in 3D bioprinted scaffolds, particularly when transitioning from single filament to sheet-like structures.

The correlation between cellular alignment at the center of the single line (defined as Line C) and the margins of the infill part (scaffold edge)

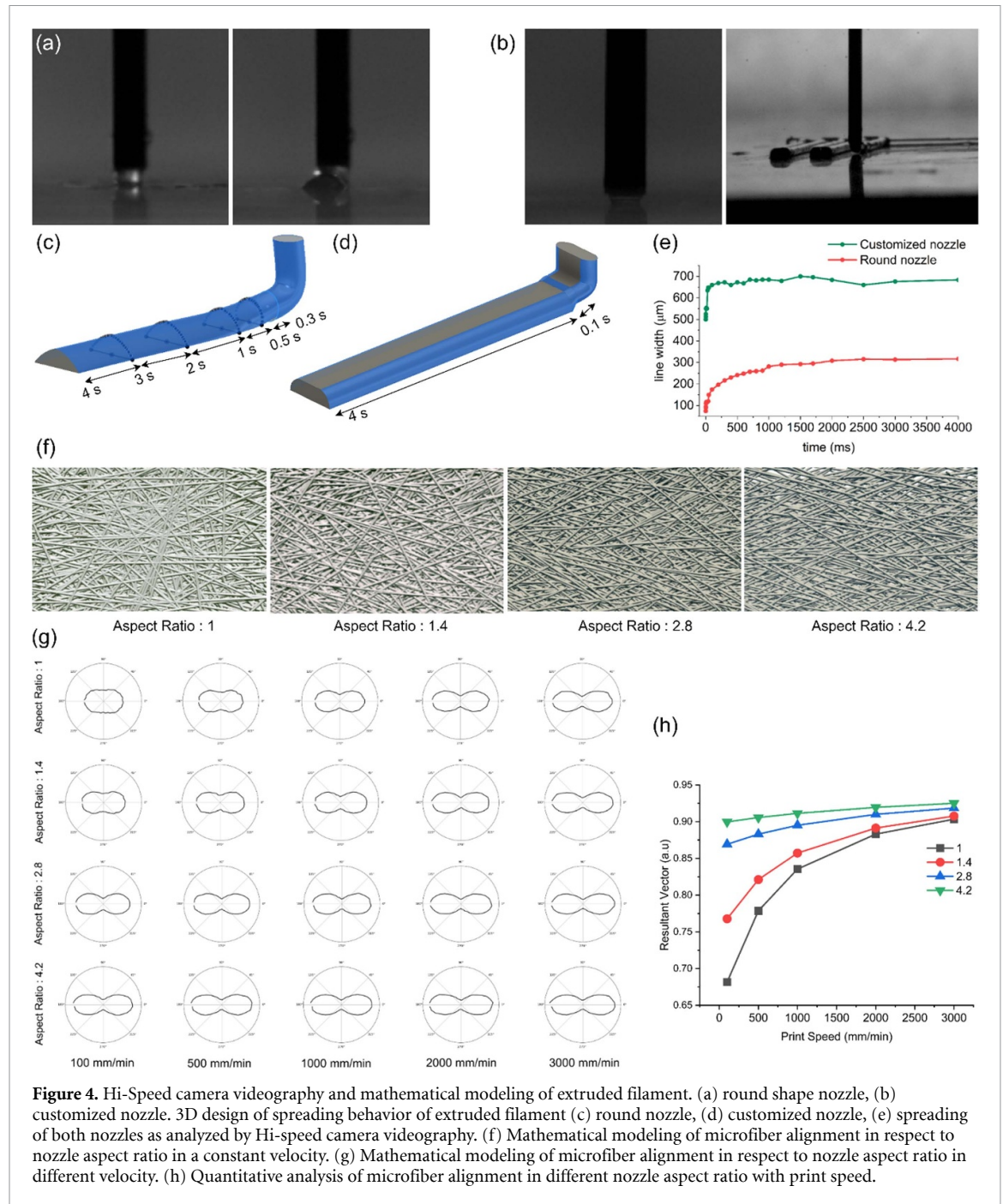


Figure 4. Hi-Speed camera videography and mathematical modeling of extruded filament. (a) round shape nozzle, (b) customized nozzle. 3D design of spreading behavior of extruded filament (c) round nozzle, (d) customized nozzle, (e) spreading of both nozzles as analyzed by Hi-speed camera videography. (f) Mathematical modeling of microfiber alignment in respect to nozzle aspect ratio in a constant velocity. (g) Mathematical modeling of microfiber alignment in respect to nozzle aspect ratio in different velocity. (h) Quantitative analysis of microfiber alignment in different nozzle aspect ratio with print speed.

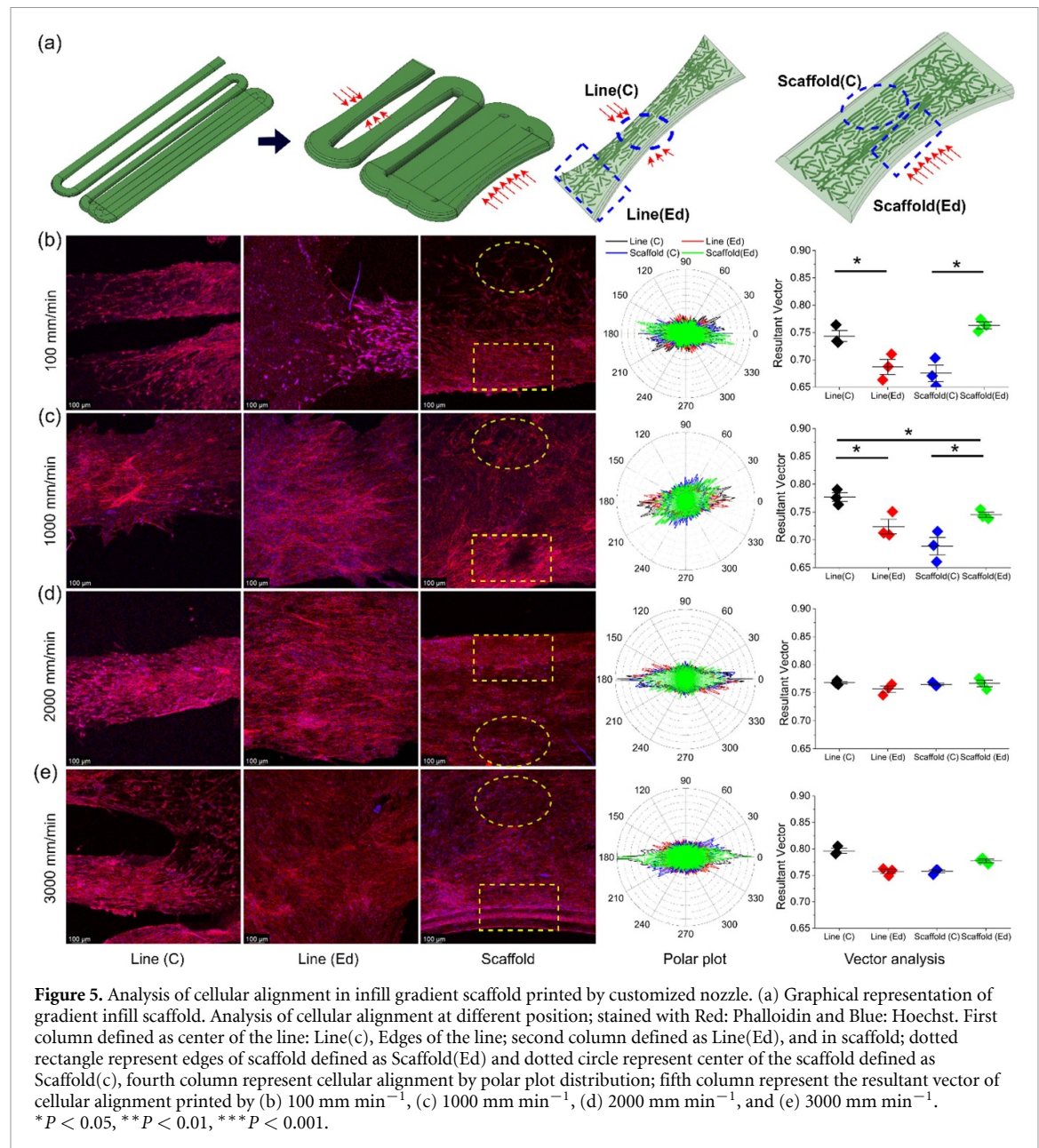
observed across all scaffolds is indeed remarkable. This alignment consistency suggests a common mechanism, likely influenced by hydrogel spreading during printing and subsequent contraction during the post-printing incubation period. Despite the reduced spreading and contraction with our customized nozzle, the edge effect appears to significantly contribute to improved alignment (figures 5(b)–(d)). This observation is corroborated by SEM images, which reveal more pronounced fiber alignment at the center of the line and less alignment towards the edges.

Our objective in fabricating sheet-like scaffolds is to achieve maximum alignment, particularly in regions with high infill percentages. Previous findings

indicate that to attain a scaffold with cellular alignment, printing speeds of at least 2000 mm min^{-1} are necessary.

3.5. Biofabrication of aligned sheet by customized nozzle

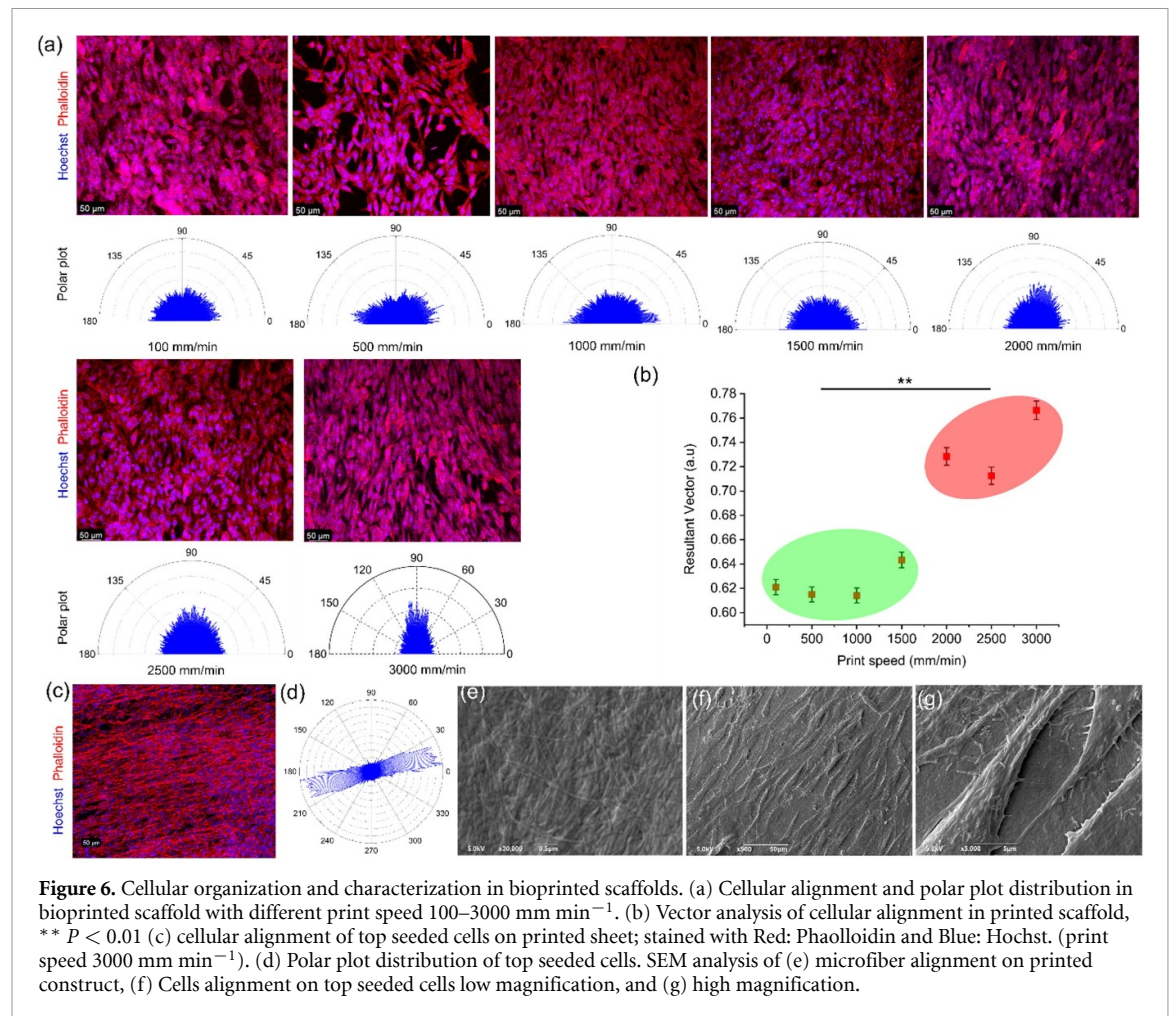
To validate our findings regarding microfiber and cell alignment in single extruded filaments and gradient infill scaffolds, we proceeded to bioprint cell-encapsulated scaffolds. Guided by the rheological behavior of our hydrogel, we selected print speeds ranging from 100 to 3000 mm min^{-1} to examine ECM fiber orientation and subsequent cellular organization in response to nozzle geometry and applied shear stress. Utilizing NIH 3T3 mouse fibroblast



cells, we bioprinted a $15 \times 10 \text{ mm}^2$ scaffold with 100% infill across various print speeds. These scaffolds underwent a 7 day culture period with media changes every other day. Subsequently, we quantified cellular orientation relative to the print direction through confocal microscopy imaging, following staining with Phalloidin and Hoechst. Our results revealed that encapsulated cells exhibit orientation along the printing path when printed at specific speeds or shear stress levels. Quantitative assessment of cellular orientation demonstrated nearly uniaxial alignment along the fiber axis. While a single line with cellular alignment could be achieved at lower print speeds, a print speed of 2000 mm min^{-1} or higher was required to fabricate a scaffold with our customized nozzle (figures 6(a) and (b)). At print speeds ranging from 100 to 1000 mm min^{-1} , the average resultant vector was measured at 0.633 ± 0.03 . Remarkably, this

value significantly increased to 0.713 ± 0.02 at print speeds between 2000 and 3000 mm min^{-1} .

Moreover, we have showcased the potential of our proposed biofabrication strategy to create a scaffold with aligned ECM microfibers and seeded cells oriented in the desired direction. By optimizing printing parameters and the direction of nozzle movement, we can directly control cellular orientation by aligning the microfibers. To further scrutinize cellular orientation on an aligned sheet, we printed a sheet-like scaffold without cells at a printing speed of 3000 mm min^{-1} and subsequently seeded NIH 3T3 mouse fibroblast cells on the top surface. The seeded cells exhibited high alignment with the orientation of the microfibers, resulting in the formation of a highly aligned cellular scaffold with a resultant vector of 0.8542 ± 0.01 . As illustrated in figures 6(c) and (d), the top-seeded cells align themselves along the



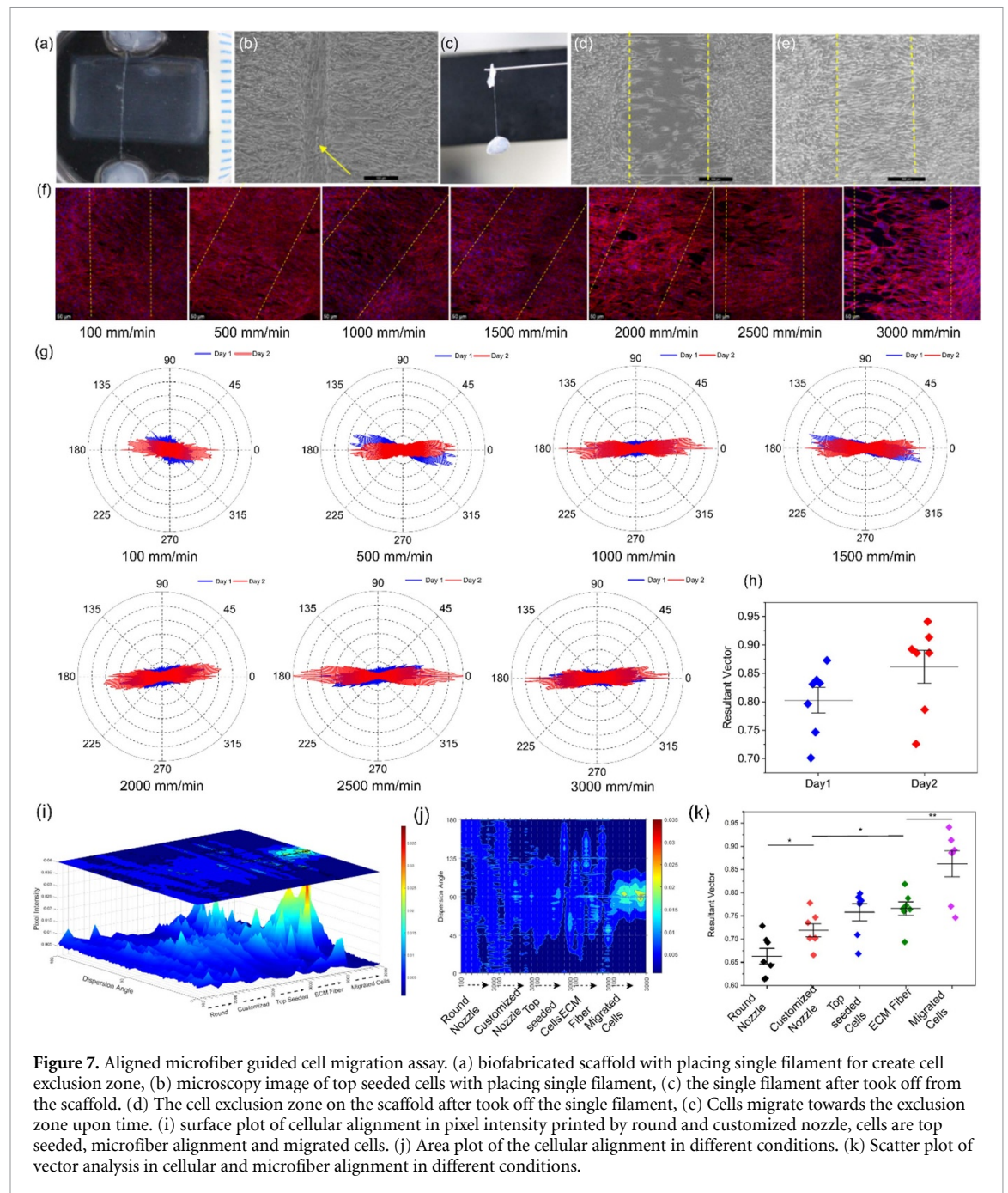
direction of the ECM microfibers, with the movement of pseudopodia also aligning towards the fiber direction. Examination of the top surface of the aligned sheet via SEM imaging further confirmed the uniaxial alignment of ECM microfibers towards the printing direction. Additionally, cross-validation of cellular alignment on top-seeded cells through SEM analysis revealed alignment towards the orientation of microfibers (figures 6(e)–(g)).

3.6. Aligned ECM fiber guided cell migration

For our final demonstration, we delved into analyzing the impact of ECM fiber alignment on cell migration. Matrix alignment is widely acknowledged as a primary factor influencing cell orientation and migration, pivotal for numerous biological and physiological processes [35–42]. Alongside the aligned fibers present within the core and surface of our scaffold, we conducted a thorough evaluation of surface topography to gauge its influence on cell migration.

Utilizing a cell exclusion zone-based migration assay atop the printed scaffold, we assessed the trajectory of cell migration. To create the cell exclusion zone, a single printed line with the same

hydrogel was placed and subsequently removed after the top-seeded cells reached confluence (figures 7(a)–(c)). Bright field microscopy images were captured and analyzed for alignment (figures 7(d), (e) and S13). After covering the cell exclusion zone, the sample was fixed, stained with Phalloidin and Hoechst, and imaged using a confocal microscope (figure 7(f)). Our observations unveiled that on the first day, cells-initiated migration towards the cell exclusion zone in a manner aligned with the printing direction. As the print speed increased, the alignment of migrated cells became more conspicuous, particularly noticeable in samples printed at a speed of 1000 mm min⁻¹ or higher (figure 7(g)). This alignment trend persisted across all scaffolds, and by the second day, the cell exclusion zone was fully populated with migrated cells. Analysis of migration pathways depicted in bright field images from day 1 and day 2 distinctly illustrated that cells migrated along the path set by the initial cells on day 1, coinciding with the printing path. Further, we analyzed the resultant vector of cell alignment for both day 1 and day 2 images, revealing no significant difference between groups in terms of alignment (figure 7(h)). The presence of an aligned ECM matrix underneath facilitated



organized cell migration, holding profound implications for various physiological functions.

Though the aligned microfiber guided the cellular alignment, the cells responded to it differently and were able to alter the degree of alignment. For example, the degree of alignment increased when the cells were top seeded rather than encapsulated in three dimensions, and it increased even farther when the cells were allowed to migrate on top of the scaffold (figures 7(i)–(k)).

3.7. Fabrication a cardiac tissue patch with aligned H9C2 encapsulated cells

The fiber-based biofabrication method holds promise for creating biomimetic scaffolds, especially where

anisotropy plays a crucial role in functionalizing tissue scaffolds. By mimicking the fibrillar structural organization of tissues, this approach can elicit synchronous cellular behavior conducive to tissue function, such as in myocardial tissue regeneration [43–45]. Myocardial tissue primarily comprises collagen (70%), with fibrillar collagen type-1 constituting a significant portion (85%) [46–48].

In this proof-of-concept demonstration, we utilized our collagen-1 rich dermal matrix [26, 49] and leveraged the fibrillogenesis kinetics to fabricate an aligned cardiac tissue patch embedded with H9C2 rat embryonic cardiomyocyte cells. Employing our customized nozzle, we bioprinted the cell-encapsulated bioink at a printing speed of 3000 mm min^{-1} into

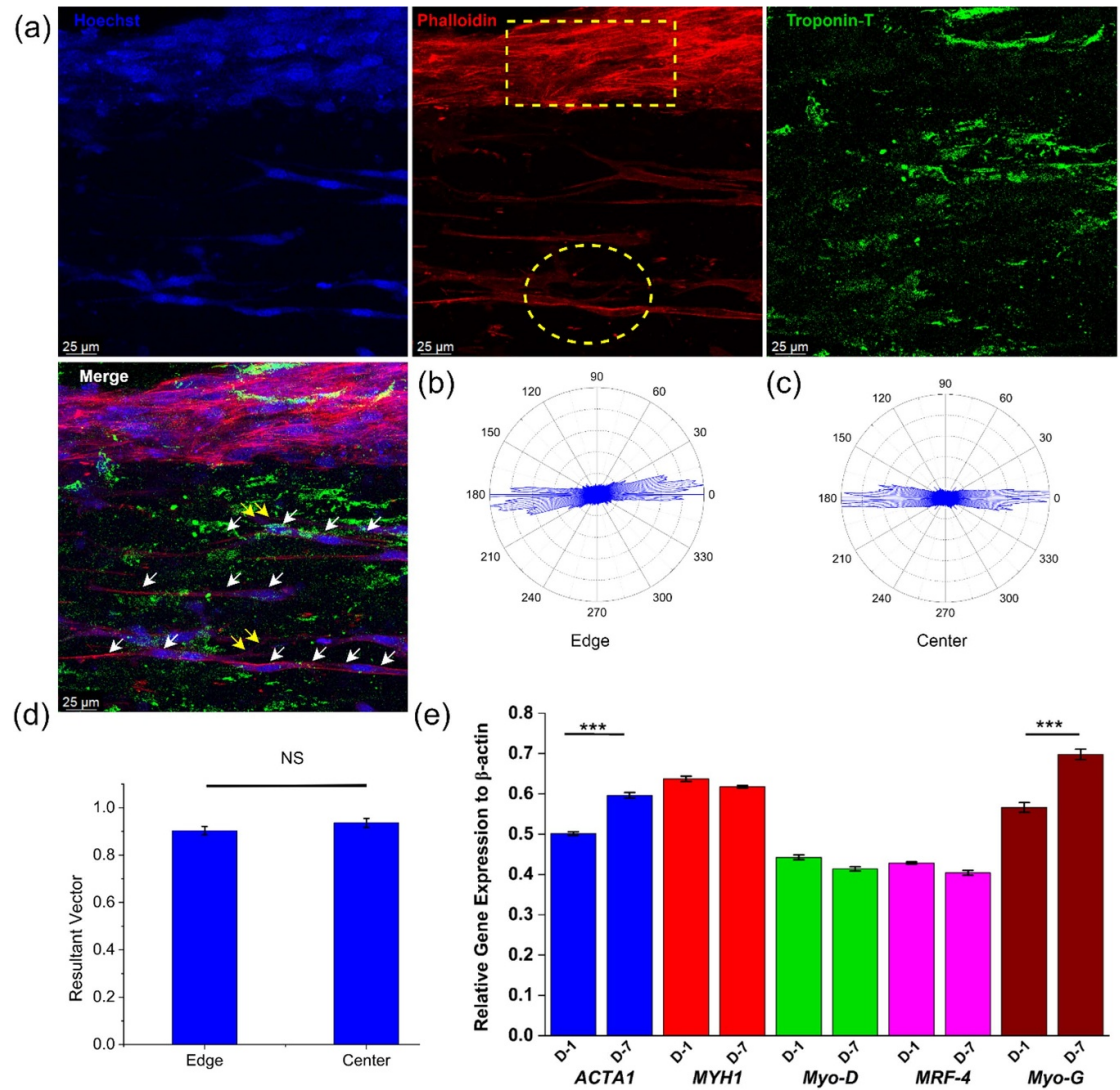


Figure 8. Application of microfiber aligned bioprinted scaffold for heart tissue engineering. (a) Morphology analysis in H9C2 myoblast encapsulated bioprinted scaffold stained with Phalloidin, and expression of Troponin-T of the scaffold, in the merge image white arrow indicate multicellular myotube formation, and yellow arrow indicated intercalated disc formation. (b) Polar plot of cellular orientation and edge of the scaffold, and (c) in center. (d) Vector analysis of cellular alignment of the scaffold at the edge and center of the scaffold. (e) Gene expression analysis by RT-PCR in bioprinted scaffold in compared to day 1 and day7, the expression level of different gene represented as relative expression of house-keeping gene β -actin, *** $P < 0.001$.

a scaffold measuring $15 \times 10 \text{ mm}^2$. After fabricating and culturing the scaffold for 7 d, we analyzed its maturation and alignment using confocal microscopy, with Phalloidin and Hoechst staining revealing intriguing findings. Confocal imaging showcased extensive stretching and alignment of cells within the scaffold, indicating a specific organization pattern (figure 8(a)). Notably, both cellular microfilaments and nuclei displayed alignment and stretching, suggesting a guided orientation that likely influences their cellular and metabolic activities. Early indications from research suggest that H9C2 cells can differentiate into myoblasts when organized in a specific manner and interact with one another. On the seventh day of culture, notable developments were observed, including the emergence of multicellular myotubes (figure 8(a), merge image white arrow).

These myotubes eventually formed interconnected tubes, reminiscent of intercalated disc characteristic of heart muscle (figure 8(a), merge image yellow arrow). This demonstration highlights the potential of our biofabrication approach in guiding cellular organization and maturation toward tissue-specific architecture, holding promise for applications in cardiac tissue engineering and regenerative medicine.

We specifically selected Troponin-T, an active contractile marker, to assess cellular activity and the capability to construct a cardiac tissue patch. Impressively, our efforts yielded a cardiac tissue patch harboring implanted cardiomyocytes expressing Troponin-T and exhibiting contractile behavior akin to in vivo conditions (figure 8(a)). Despite some contractions observed at the scaffold edges (figures 8(b)–(d)), the cells demonstrated high alignment and

interconnectivity, forming a contractile tissue construct characterized by Troponin-T expression. To further corroborate the differentiation of H9C2 cells into myotubes and their progression toward functional cardiac tissue patch formation, we examined the messenger RNA (mRNA) expression of various transcription factors crucial for muscle development. Of particular interest was ACTA1, the gene encoding α -actin, which plays a pivotal role in sarcomere formation during muscle contraction. Our bioprinted constructs exhibited a significant increase in α -actin expression on day 7 compared to day 1, suggesting ongoing maturation and development towards a functional cardiac tissue patch.

Further, we proceeded to assess the expression level of MYH1, responsible for muscle contraction along with actin, which has been consistently expressed since day 1 and exhibited minimal change by day 7 [50]. In examining the potential influence of cellular alignment on the myogenic lineage of developed cardiomyoblasts, we investigated the expression of early differentiation factor (Myo-D) and late differentiation factors (MRF-4 and Myo-G) in bioprinted aligned cardiomyocytes [51–53]. Comparing day 7 to day 1, we observed no notable difference in the expression of the early differentiation marker Myo-D. Concerning the late differentiation markers, there was no significant alteration in MRF-4 expression, whereas Myo-G expression showed a substantial upregulation on day 7 compared to day 1 (figure 8(e)). Consequently, our findings suggest that cellular alignment within the bioprinted scaffold may promote H9C2 cell differentiation towards a cardiac tissue patch.

Altogether, the study demonstrated the fabrication of a cardiac tissue patch using a collagen-rich dermal matrix and embedded cardiomyocyte cells. Confocal microscopy revealed extensive cellular alignment and stretching within the scaffold, with the development of multicellular myotubes resembling features of heart tissue. The expression of Troponin-T indicated contractile activity, while mRNA analysis showed upregulation of Myo-D, suggesting differentiation towards cardiac tissue. Cellular alignment appeared to promote differentiation, highlighting the potential for engineered cardiac patches in regenerative medicine.

The current mathematical modeling and experimental results demonstrate the successful fabrication of a tissue analog with aligned microfibers and encapsulated cells oriented in the direction of printing path. However, given that the nozzle has an aspect ratio greater than 1 and there is no control over the angular rotation of the extrusion nozzle, this may impact the ability to fabricate circular constructs with uniform cellular alignment. Hence, the present approach is effective for linear nozzle movement but may not be

suitable for applications requiring angular or complex patterns, such as zigzag or wavy configurations.

4. Conclusion

We have developed a robust biofabrication method for creating tissue patches with precisely aligned microfibers and encapsulated cells. Leveraging caprine dermal tissue rich in Type-I fibrillar collagen, we have fine-tuned printing parameters to optimize alignment through extrusion-based bioprinting. Our approach considers factors like nozzle shape, shear stress, and extensional force, revealing their critical roles in controlling microfiber alignment and cellular orientation within the construct. By customizing the nozzle design, we have minimized post-printing filament spreading, achieving comparable alignment at lower print speeds. Computational modeling validated our experimental findings, highlighting the significance of shear stress and nozzle aspect ratio in enhancing alignment. Furthermore, our method's versatility was demonstrated by successfully crafting a cardiac tissue patch with aligned ECM microfibers and encapsulated cardiomyocyte cells, showcasing expression of key markers for myotube formation and contractility. Overall, our approach offers a promising avenue for creating tissue-specific microenvironmental architectures with precise cell and ECM fiber alignment. Nevertheless, this method's adaptability extends beyond cardiac tissue, making it applicable for engineering a wide range of anisotropic tissue types.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgment

The authors acknowledge the funding agency for this work is CSIR-UGC, MHRD Ministry of India.

Conflict of interest

Authors declare there have no conflict of interest.

ORCID iDs

Ashis Kumar Bera  <https://orcid.org/0009-0009-8372-6993>

Mohd Suhail Rizvi  <https://orcid.org/0000-0002-4130-4671>

Vijayasankar KN  <https://orcid.org/0000-0002-3096-4412>

Falguni Pati  <https://orcid.org/0000-0002-3588-1800>

References

- [1] Bhatia M 2010 Microenvironment mimicry *Science* **329** 1024–5
- [2] Black L D, Meyers J D, Weinbaum J S, Shvelidze Y A and Tranquillo R T n.d. Cell-induced alignment augments twitch force in fibrin gel-based engineered myocardium via gap junction modification (available at: www.liebertonline.com)
- [3] Wang J H C 2006 Mechanobiology of tendon *J. Biomech.* **39** 1563–82
- [4] Shah A, Brugnano J, Sun S, Vase A and Orwin E 2008 The development of a tissue-engineered cornea: biomaterials and culture methods *Pediatr. Res.* **63** 535–44
- [5] Saw T B, Doostmohammadi A, Nier V, Kocgozlu L, Thampi S, Toyama Y, Marcq P, Lim C T, Yeomans J M and Ladoux B 2017 Topological defects in epithelia govern cell death and extrusion *Nature* **544** 212–6
- [6] Turiv T, Krieger J, Babakhanova G, Yu H, Shiyankovskii S V, Wei Q-H, Kim M-H and Lavrentovich O D 2020 Topology control of human fibroblast cells monolayer by liquid crystal elastomer (available at: www.science.org)
- [7] Cheng Q et al 2023 Decellularized Scaffolds with double-layer aligned microchannels induce the oriented growth of bladder smooth muscle cells: toward urethral and ureteral reconstruction *Adv. Healthcare Mater.* **12** e2300544
- [8] House A, Cornick J, Butt Q and Guvendiren M 2023 Elastomeric platform with surface wrinkling patterns to control cardiac cell alignment *J. Biomed. Mater. Res. A* **111** 1228–42
- [9] Mirbagheri M, Adibnia V, Hughes B R, Waldman S D, Banquy X and Hwang D K 2019 Advanced cell culture platforms: a growing quest for emulating natural tissues *Mater. Horiz.* **6** 45–71
- [10] Tan J, Shen H and Saltzman W M 2001 Micron-scale positioning of features influences the rate of polymorphonuclear leukocyte migration *Biophys. J.* **81** 2569–79
- [11] Costa K D, Lee E J and Holmes J W 2003 Creating alignment and anisotropy in engineered heart tissue: role of boundary conditions in a model three-dimensional culture system *Tissue Eng.* **9** 567–77
- [12] Guyette J P et al 2016 Bioengineering human myocardium on native extracellular matrix *Circ. Res.* **118** 56–72
- [13] Alapan Y, Younesi M, Akkus O and Gurkan U A 2016 Anisotropically stiff 3D micropillar niche induces extraordinary cell alignment and elongation *Adv. Healthcare Mater.* **5** 1884–92
- [14] Sriram M, Priya S, Mahajan A and Katti D S 2024 Directing ligament-mimetic bi-directional cell organization in scaffolds through zone-specific microarchitecture for ligament tissue engineering *Biofabrication* **16** 025015
- [15] Luo Y and Shoichet M S 2004 A photolabile hydrogel for guided three-dimensional cell growth and migration *Nat. Mater.* **3** 249–54
- [16] Lee S H, Moon J J and West J L 2008 Three-dimensional micropatterning of bioactive hydrogels via two-photon laser scanning photolithography for guided 3D cell migration *Biomaterials* **29** 2962–8
- [17] Justin A W et al 2023 Densified collagen tubular grafts for human tissue replacement and disease modelling applications *Biomater. Adv.* **145** 213245
- [18] Zhu M et al 2019 In vivo engineered extracellular matrix scaffolds with instructive niches for oriented tissue regeneration *Nat. Commun.* **10** 4620
- [19] Aubin H, Nichol J W, Hutson C B, Bae H, Sieminski A L, Cropek D M, Akhyari P and Khademhosseini A 2010 Directed 3D cell alignment and elongation in microengineered hydrogels *Biomaterials* **31** 6941–51
- [20] Zhang Y, O'Mahony A, He Y and Barber T 2024 Hydrodynamic shear stress' impact on mammalian cell properties and its applications in 3D bioprinting *Biofabrication* **16** 22003
- [21] Ahrens J H, Uzel S G, Skylar-Scott M, Mata M M, Lu A, Kroll K T and Lewis J A 2022 Programming cellular alignment in engineered cardiac tissue via bioprinting anisotropic organ building blocks *Adv. Mater.* **34** 2200217
- [22] Samandari M, Alipanah F, Majidzadeh-A K, Alvarez M M, Trujillo-De Santiago G and Tamayol A 2021 Controlling cellular organization in bioprinting through designed 3D microcompartmentalization *Appl. Phys. Rev.* **8** 021404
- [23] Stewart M H, Bendall S C and Bhatia M 2008 Deconstructing human embryonic stem cell cultures: niche regulation of self-renewal and pluripotency *J. Mol. Med.* **86** 875–86
- [24] Pati F, Jang J, Ha D H, Won Kim S, Rhie J W, Shim J H, Kim D H and Cho D W 2014 Printing three-dimensional tissue analogues with decellularized extracellular matrix bioink *Nat. Commun.* **5** 3935
- [25] Badyal S F 2007 The extracellular matrix as a biologic scaffold material *Biomaterials* **28** 3587–93
- [26] Bera A K, Sriya Y and Pati F 2022 Formulation of dermal tissue matrix bioink by a facile decellularization method and process optimization for 3D bioprinting toward translation research *Macromol. Biosci.* **22** 2200109
- [27] Ness C, Seto R and Mari R 2022 The physics of dense suspensions *Annu. Rev. Condens. Matter Phys.* **13** 97–117
- [28] Kim H, Jang J, Park J, Lee K P, Lee S, Lee D M, Kim K H, Kim H K and Cho D W 2019 Shear-induced alignment of collagen fibrils using 3D cell printing for corneal stroma tissue engineering *Biofabrication* **11** 035017
- [29] Schwab A, Levato R, D'Este M, Piluso S, Eglin D and Malda J 2020 Printability and shape fidelity of bioinks in 3D bioprinting *Chem. Rev.* **120** 11028–55
- [30] Gillispie G, Prim P, Copus J, Fisher J, Mikos A G, Yoo J J, Atala A and Lee S J 2020 Assessment methodologies for extrusion-based bioink printability *Biofabrication* **12** 022003
- [31] Fallahi A et al 2020 Customizable composite fibers for engineering skeletal muscle models *ACS Biomater. Sci. Eng.* **6** 1112–23
- [32] Kim W J, Kim M and Kim G H 2018 3D-printed biomimetic scaffold simulating microfibril muscle structure *Adv. Funct. Mater.* **28** 1800405
- [33] Nerger B A, Brun P T and Nelson C M 2019 Microextrusion printing cell-laden networks of type I collagen with patterned fiber alignment and geometry *Soft Matter* **15** 5728–38
- [34] Kim W J and Kim G H 2020 3D bioprinting of functional cell-laden bioinks and its application for cell-alignment and maturation *Appl. Mater. Today* **19** 100588
- [35] Tran V D and Kumar S 2021 Transduction of cell and matrix geometric cues by the actin cytoskeleton *Curr. Opin. Cell Biol.* **68** 64–71
- [36] SenGupta S, Parent C A and Bear J E 2021 The principles of directed cell migration *Nat. Rev. Mol. Cell Biol.* **22** 529–47
- [37] Renkawitz J et al 2019 Nuclear positioning facilitates amoeboid migration along the path of least resistance *Nature* **568** 546–50
- [38] Yamada K M and Sixt M 2019 Mechanisms of 3D cell migration *Nat. Rev. Mol. Cell Biol.* **20** 738–52
- [39] Qin L, Yang D, Yi W, Cao H and Xiao G 2021 Roles of leader and follower cells in collective cell migration *Mol. Biol. Cell* **32** 1267–72
- [40] Wang W Y, Pearson A T, Kutys M L, Choi C K, Wozniak M A, Baker B M and Chen C S 2018 Extracellular matrix alignment dictates the organization of focal adhesions and directs uniaxial cell migration *APL Bioeng.* **2** 046107
- [41] Sun X, Driscoll M K, Guven C, Das S, Parent C A, Fourkas J T and Losert W 2015 Asymmetric nanotopography biases cytoskeletal dynamics and promotes unidirectional cell guidance *Proc. Natl Acad. Sci. USA* **112** 12557–62
- [42] Kievit F M, Cooper A, Jana S, Leung M C, Wang K, Edmondson D, Wood D, Lee J S H, Ellenbogen R G and Zhang M 2013 Aligned chitosan-polycaprolactone polyblend nanofibers promote the migration of glioblastoma cells *Adv. Healthcare Mater.* **2** 1651–9

- [43] Zhang W et al 2019 Scaffold with micro/macro-architecture for myocardial alignment engineering into complex 3D cell patterns *Adv. Healthcare Mater.* **8** e1901015
- [44] Tijore A, Irvine S A, Sarig U, Mhaisalkar P, Baisane V and Venkatraman S 2018 Contact guidance for cardiac tissue engineering using 3D bioprinted gelatin patterned hydrogel *Biofabrication* **10** 25003
- [45] Fan T, Wang S, Jiang Z, Ji S, Cao W, Liu W, Ji Y, Li Y, Shyh-Chang N and Gu Q 2021 Controllable assembly of skeletal muscle-like bundles through 3D bioprinting *Biofabrication* **14** 015009
- [46] Badylak S F, Taylor D and Uygun K 2011 Whole-organ tissue engineering: decellularization and recellularization of three-dimensional matrix scaffolds *Annu. Rev. Biomed. Eng.* **13** 27–53
- [47] Bashey R I, Martinez-Hernandez A and Jimenez S A 1990 Isolation, characterization, and localization of cardiac collagen type VI: associations with other extracellular matrix components *Circ. Res.* **70** 1006–17
- [48] Li H, Bao M and Nie Y 2021 Extracellular matrix-based biomaterials for cardiac regeneration and repair *Card. Fail. Rev.* **26** 1231–48
- [49] Bera A K and Pati F 2023 3D bioprinting of skin tissue model *Skin 3-D Models and Cosmetics Toxicity* ed A B Pant, A Dwivedi, R S Ray, A Tripathi, A K Upadhyay and S Poojan pp 83–104 (Springer)
- [50] Boonen K J M, Langelaan M L P, Polak R B, van der Schaft D W J, Baaijens F P T and Post M J 2010 Effects of a combined mechanical stimulation protocol: value for skeletal muscle tissue engineering *J. Biomech.* **43** 1514–21
- [51] Kopantseva E E and Belyavsky A V 2016 Key regulators of skeletal myogenesis *Mol. Biol.* **50** 195–222
- [52] Karalaki M, Fili S, Philippou A and Koutsilieris M 2009 Muscle regeneration: cellular and molecular events *In Vivo* **23** 779–96
- [53] Zevolis E, Philippou A, Moustogiannis A, Chatzigeorgiou A and Koutsilieris M 2021 Optimizing mechanical stretching protocols for hypertrophic and anti-apoptotic responses in cardiomyocyte-like H9C2 cells *Mol. Biol. Rep.* **48** 645–55